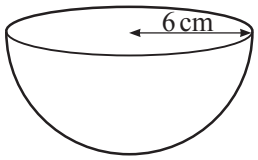
4 **(a)**

NOT TO SCALE



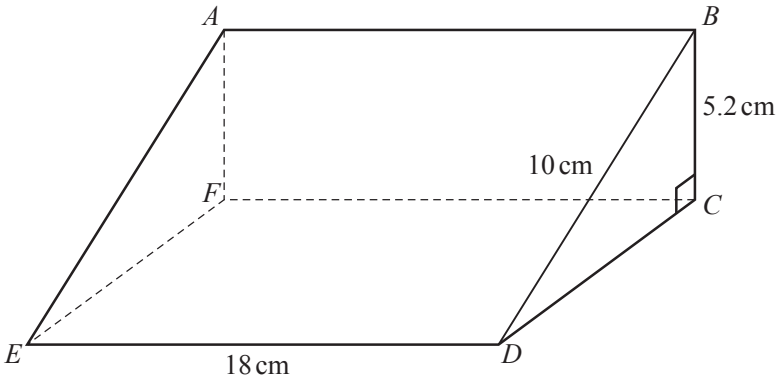
The diagram shows a hemisphere with radius 6 cm.

Calculate the volume.
Give the units of your answer.

[The volume, V, of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.]

.... [3]

(b)



NOT TO SCALE

The diagram shows a prism *ABCDEF*.
The cross-section is a right-angled triangle *BCD*.
$BD = 10$ cm, $BC = 5.2$ cm and $ED = 18$ cm.

(i) **(a)** Work out the volume of the prism.

..... cm^3 [6]

(b) Calculate angle *BEC*.

Angle *BEC* = ... [4]

(ii) The point *G* lies on the line *ED* and *GD* = 7 cm.

Work out angle *BGE*.

Angle *BGE* = ... [3]